INDIVIDUAL WEEKLY PROJECT REPORTS

Student: Annalakshmi R

Role: Data Preprocessing and Exploratory Data Analysis (EDA)
Project Title: Flight Delay Analysis and Visualization

WEEK 6 – Advanced Exploratory Analysis and Feature Preparation

During the sixth week, I continued working on the analysis of flight delay data and focused on understanding how relevant variables could be prepared for further investigation. I explored the importance of selecting appropriate features and organizing the available information for statistical analysis.

I examined the relationship between different flight-related attributes and considered how variables such as departure delay, arrival delay, scheduled timings, and elapsed time could contribute to the analysis.

I also understood the importance of creating meaningful categories from existing information. For example, time-related

variables can be grouped into suitable periods to make comparisons easier. Such preparation can help identify whether certain time periods show different delay patterns.

I coordinated with the team to ensure that the variables selected for further analysis were consistent with the project's objectives. I also reviewed the prepared data to understand whether any additional preprocessing was necessary.

Another part of my work involved examining the results of exploratory analysis and identifying observations that could be included in the report and dashboard.

Outcome and Learning:

This week helped me understand the connection between exploratory data analysis and feature engineering. I learned that preparing meaningful variables can make further analysis more effective and easier to interpret.